Four summands: moment paths and active collisions

28 October 2026

A linear moment theorem does not compress an iid four-fold convolution. The counterexample below occurs inside the bounded moment and threshold box used by the current four-summand reduction. A contact inequality does supply a different restriction: apart from at most two points, every atom of a finite maximizing law must occur in an active four-sum collision. This is a structural reduction with explicit remaining cases, not a proof of the four-summand Berry–Esseen bound.

1 An exact obstruction to moment-extreme compression

Let Z_θ have support $\{-2,-1,0,1,2\}$ and probability vector

$$\left(\frac{1}{20}+\theta,\ \frac{1}{5}-2\theta,\ \frac{1}{2},\frac{1}{5}+2\theta,\ \frac{1}{20}-\theta\right), \qquad -\frac{1}{20}\le\theta\le\frac{1}{20}. \tag{1}$$

All these laws have

$$\mathbb{E}Z_\theta=0, \qquad \mathbb{E}Z_\theta^2=\frac{4}{5}, \qquad \mathbb{E}|Z_\theta|^3=\frac{6}{5}.$$

Consequently $X_\theta=(\sqrt{5}/2)Z_\theta$ is standardized and has $\rho=3\sqrt{5}/4<46/25$. Its support is contained in $[-\sqrt{5},\sqrt{5}]\subset(-4.032,4.032)$, and $\beta=\mathbb{E}[X_\theta|X_\theta|]=-5\theta$, so $|\beta|\le 1/4$. The fixed standardized raw sum threshold corresponding to $\sum_{j=1}^4 Z_{\theta,j}\le 1$ is $H=\sqrt{5}/2$.

Proposition 1. *On the set of probability laws supported on these five raw points with these three moments, the four-summand closed CDF at threshold* 1 *attains its maximum strictly inside the moment segment. In particular neither extreme law maximizes it. The same assertion holds for the standardized fixed-target functional*

$$2\{\mathbb{P}(X_{\theta,1}+\cdots+X_{\theta,4}\le H)-\Phi(H/2)\}-c\rho$$

for every fixed c, because its other terms are constant on the segment.

Proof. The matrix with rows $1, z, z^2, |z|^3$ on the five support points has rank four. Its nullspace is spanned by $(1,-2,0,2,-1)$; nonnegativity gives exactly the interval in (1). Hence the two endpoints are the only extreme points of this moment polytope. Direct convolution, with all threshold atoms included, gives

$$\begin{aligned} F(\theta)&:=\mathbb{P}(Z_{\theta,1}+\cdots+Z_{\theta,4}\le 1)\\ &=\frac{25719}{32000}-\frac{103}{2000}\theta-\frac{483}{200}\theta^2-\frac{123}{5}\theta^3+11\theta^4. \end{aligned} \tag{2}$$

In particular

$$F(0)=\frac{25719}{32000}, \quad F(-1/20)=\frac{4017}{5000}, \quad F(1/20)=\frac{7921}{10000}.$$

The two strict differences are 51/160000 and 1859/160000. Continuity gives a maximizer, and neither endpoint can be one. More explicitly,

$$F'(\theta) = \frac{(20\theta + 1)(4400\theta^2 - 7600\theta - 103)}{2000}.$$

The second factor decreases on the entire interval and changes sign between $-1/40$ and $-1/100$. Thus the unique maximum is at

$$\theta_* = \frac{190 - \sqrt{37233}}{220} \in (-1/40, -1/100).$$

The standardization and fixed-target assertions follow from the common moments and common threshold. □

This is not a counterexample to the Berry–Esseen conjecture. It refutes an intermediate extremal principle. It even has a positive closed normal discrepancy: $F(0) > 4/5$, whereas $\Phi(\sqrt{5}/4) \le 1/2 + \sqrt{5}/10 < 73/100$. The example does not preserve the signed third moment: $\mathbb{E}X_\theta^3 = -(15\sqrt{5}/2)\theta$. A reduction imposing that additional constraint would require its own argument. Nor does the example assert that its interior maximizer is a global fixed-target maximizer over all probability laws.

2 The correct finite moment-path calculation

For finite signed measures define

$$\mathcal{A}_H(\mu_1, \mu_2, \mu_3, \mu_4) = \int \mathbf{1}_{\{x_1+x_2+x_3+x_4 \le H\}}\, d\mu_1\, d\mu_2\, d\mu_3\, d\mu_4.$$

This is symmetric and multilinear. The notation $\mathcal{A}_H(P^{4-j}, \nu^j)$ records how many arguments are of each type.

Lemma 2 (A complete one-dimensional replacement)**.** *Let P be a probability measure with finite absolute third moment and let ν be a finite signed measure with $\int (1 + |x|^3)\, d|\nu|(x) < \infty$. Assume $P + t\nu$ is a nonnegative probability measure for $t \in [a, b]$. Suppose*

$$\int (1, x, x^2, |x|^3)\, d\nu(x) = 0.$$

The mean, variance and absolute third moment remain fixed, and

$$F^{*4}_{P+t\nu}(H) = \sum_{j=0}^{4} \binom{4}{j} t^j \mathcal{A}_H(P^{4-j}, \nu^j). \tag{3}$$

Its maximum is attained at an endpoint or an interior zero of the cubic derivative. Thus an exact derivative-root check, or a whole-interval quartic majorant, is sufficient. Checking only the endpoints is not.

Proof. The moments are linear. Expand the four product factors, grouping equal terms by symmetry. The event stays fixed, including its boundary; no continuity of its indicator is used. The remaining assertion is the ordinary extreme-value theorem and Fermat's theorem for a polynomial. □

A useful necessary test for a local maximum is also immediate. If both signs of t are feasible near zero, then

$$\mathcal{A}_H(P^3, \nu) = 0, \qquad \mathcal{A}_H(P^2, \nu^2) \le 0.$$

If the second quantity vanishes, then also $\mathcal{A}_H(P, \nu^3) = 0$ and $\mathcal{A}_H(\nu^4) \le 0$. For $P = \sum_{j=1}^m p_j \delta_{x_j}$ with $p_j > 0$, write $\nu = \sum_j v_j \delta_{x_j}$ and

$$K_{ij} = P^{*2}((-\infty, H - x_i - x_j]).$$

The quadratic condition is $v^{\mathsf{T}} K v \le 0$ for every vector in the nullspace of the four moment rows. In other words, the compression of K to that nullspace must be negative semidefinite. This is a necessary same-law condition; it does not characterize an optimizer or replace the whole event probability. It offers an inexpensive rejection test for candidate atomic supports before any dense certificate construction.

3 At most two atoms can avoid the active collision

We now use a contact premise, which is distinct from a moment-polytope extreme-point claim.

Lemma 3 (A cubic contact potential has at most two minima)**.** *For $c > 0$ and real a, b, d, the function $q(x) = c|x|^3 + ax^2 + bx + d$ has at most two distinct local minima. If there are two, one is negative and one positive.*

Proof. If $a \ge 0$, the function is strictly convex, so has at most one local minimum. If $a < 0$, set $R = -a/(3c) > 0$. On $(-R, R)$ its second derivative $6c|x| + 2a$ is negative. On each exterior interval it is strictly positive, so the derivative has at most one zero there. A stationary point at R or $-R$ has a nonzero third derivative and is not a local minimum. These observations give at most one local minimum on each exterior interval and none elsewhere. □

Theorem 4 (Contact forces active support collisions)**.** *Let P have finite support S, with every listed atom of positive mass. Fix $H \in \mathbb{R}$, $c > 0$, and assume that for some real a, b, d*

$$q(x) := c|x|^3 + ax^2 + bx + d \;\ge\; 8P^{*3}((-\infty, H - x]), \qquad x \in \mathbb{R}, \tag{4}$$

with equality at every $x \in S$. Then at most two points of S fail to belong to an active four-sum collision. Equivalently,

$$\#\{x \in S : H - x \notin S + S + S\} \le 2.$$

If there are two such exceptional points, they have opposite signs. If exactly one unordered four-term multiset from S sums to H, and it contains r distinct points, then $\#S \le r + 2 \le 6$.

Proof. If $H - x \notin S + S + S$, the finite-support CDF $P^{*3}((-\infty, H - y])$ is constant for y in a neighborhood of x. Equality at x and (4) therefore make x a local minimum of the same function q. Apply Lemma 3. Every point that does occur in an active collision belongs to its four-term multiset. If there is only one such multiset, it contains all but at most two support points, which proves the last assertion. □

For the fixed-target objective $2\{P^{*4}((-\infty, H]) - \Phi(H/2)\} - c\rho$, the contact formula in `sections/02-fixed-target.tex` has exactly this form. For a standardized maximizing pair, put $\varphi = \phi(H/2)$, $F = P^{*4}((-\infty, H])$ and $\beta = \mathbb{E}[X|X|]$. Its potential is

$$q(x) = c|x|^3 - \frac{H\varphi + 3c\rho}{2} x^2 - (4\varphi + 3c\beta)x + 8F + \frac{H\varphi + c\rho}{2}.$$

The fixed-target contact theorem supplies (4); it is an explicit premise of the new support conclusion, not proved by a finite arithmetic check. No two- or three-summand estimate is used here. For a finite law, a maximizing threshold must itself be a four-sum atom: otherwise lowering it a little preserves the CDF and strictly improves the normal subtraction.

In particular, if distinct unordered four-term multisets of S always have distinct sums, a finite fixed-target maximizing law has at most six atoms. Conversely, a finite maximizing law with at least seven atoms must have two distinct active occupancy vectors $k, l \in \mathbb{N}^m$, with

$$\sum_j k_j = \sum_j l_j = 4, \qquad \sum_j (k_j - l_j)x_j = 0, \qquad 0 < \sum_j |k_j - l_j| \le 8.$$

Thus larger atomic candidates must lie on explicit low-integer-coefficient resonance hyperplanes. The two possible exceptional atoms can be restricted further to the local minima of the displayed cubic potential.

What remains open. This is not a universal six-atom reduction. Attainment of the unrestricted problem gives compact support, not finite support. A finite maximizing law may have several simultaneous active multisets; these resonant strata cannot be discarded as a set of parameter measure zero. Perturbing a threshold collision changes a closed event discontinuously. Nor does a finite atomic approximation of a maximizing law inherit its exact contact inequality. A complete short $n = 4$ proof must address these remaining cases, or use an original-law bound such as the retained physical ROOT4 bridge.

Scope of the bounded checks. The independent rational script verifies the moment segment, its complete quartic by both convolution and direct product enumeration, its endpoint and derivative identities, moment-null tangent expansions, and actual collision hypergraphs. It rejects endpoint-only maximization and the omission of the contact premise. These are exact controls for the lemmas, not a new acceptance of G2, G3, ROOT4 or the finite base through 41.